

Paper IV (Revised) — Active-Downcomer Composite Module (DCX): Universal Retrofit for Any Tray Type — Complete Construction, NTU/HTU Derivation, and Comparative Performance Across All Tray Families

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HyTray Series HT-04 — Detailed Engineering Disclosure

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Abstract

The DCX module converts a conventional dead-volume downcomer into a mass-transfer zone by placing a structured packing insert ($a_p = 120\text{--}200\text{ m}^2/\text{m}^3$, $\varepsilon > 0.92$) inside the downcomer and admitting a controlled vapour slipstream of 3–8% of tray vapour through a floor sparger. The module is **deck-type independent**: it attaches to the downcomer of any cross-flow tray — sieve, ripple, high-performance, or any HyTray family member — and contributes 0.10–0.25 additional theoretical stages per tray regardless of what happens on the deck above. Additionally, the insert acts as a coalescer, raising the aeration factor ϕ_{dc} from 0.5 to 0.75–0.85 and permitting downcomer inlet velocities up to 0.19–0.21 m/s, extending the weir-loading limit by $\sim 25\%$. This paper provides the complete construction drawings, NTU/HTU derivation for the counter-current packed downcomer, Nusselt film-drainage equations, and a parametric comparison showing the DCX gain applied to nine tray types simultaneously.

1 Construction and Assembly

1.1 Component Inventory

DCX Module — four fabricated components

- (1) **Structured packing insert** (52): cassettes of corrugated perforated sheet, 2 mm corrugation height, 60° crimp angle, specific area $150\text{ m}^2/\text{m}^3$, void fraction 0.95. Each cassette $\leq 400\text{ mm}$ wide $\times \leq 25\text{ kg}$ (manway passable). Cassettes drop into guide channels welded to the downcomer sidewall.
- (2) **Floor sparger plate** (54): flat plate welded to the bottom of the downcomer with drilled orifices providing the vapour slipstream. Total orifice area sized for 3–8% slipstream at $U_{sp} = 8\text{--}15\text{ m/s}$. Plate doubles as the bottom seal for the insert cassettes.
- (3) **Disengagement chimney** (58): rectangular duct ($75 \times 150\text{ mm}$ typical) welded through the froth layer, connecting the top of the insert to the vapour space above the liquid level. Chimney top has a hood to prevent splash carry-in.
- (4) **Distributor lip** (56): formed plate at the downcomer truncation (80–150 mm above the deck below) that discharges liquid as a directed sheet across 70–90% of the inlet panel width. Bolted to the downcomer apron; can be retrofit on existing truncated downcomers.

1.2 Downcomer Geometry and Slipstream Sizing

Sparger orifice sizing (maintain weep-resistance)

$$A_{sp} = \frac{\dot{V}_{slip}}{C_d U_{sp}}, \quad \dot{V}_{slip} = f_V \dot{V}_{tray}$$

Driving force balance (fixes f_V self-consistently): $\Delta P_{tray,dry} - \Delta P_{insert} = \rho_V U_{sp}^2 / 2 C_d^2$.

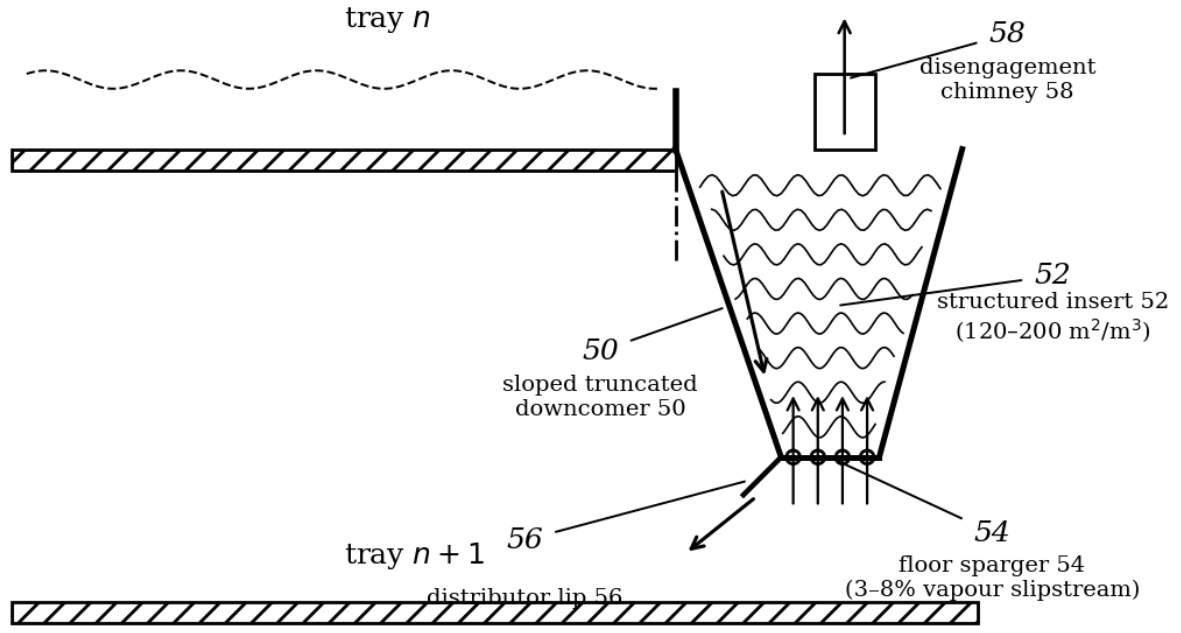


Figure 1: DCX assembly cross-section: sloped truncated downcomer (50), packing insert (52), floor sparger (54), distributor lip (56), disengagement chimney (58).

Anti-weep criterion: $U_{sp} > U_{sp,min} = \sqrt{2\rho_L g h_L / \rho_V}$ (prevents liquid rain-through sparger under low load).

1.3 Insert Flood Point and Slipstream Band

Insert flood (limits upper slipstream)

$$\text{Insert flood: } F_f = U_V \sqrt{\rho_V} \cdot U_L^{0.05} \leq F_{f,max}$$

$F_{f,max} \approx 1.8\text{--}2.5 \text{ Pa}^{0.5}$ for structured packing at $a_p = 150 \text{ m}^2/\text{m}^3$ (supplier data).

Upper slipstream limit: $f_V \leq f_{V,max}$ such that $F_f < 0.8 F_{f,max}$ at maximum weir load.

Result: claimed band 3–8% is both the mass-transfer optimum and the hydraulic safe limit.

2 Mass Transfer in the Packed Downcomer — NTU/HTU Derivation

2.1 Phase Velocities in the Downcomer

Superficial velocities inside DCX insert

$$u_{L,dc} = \frac{Q_L}{A_{dc,mean} \varepsilon}, \quad u_{V,dc} = \frac{f_V \dot{V}_{tray}}{A_{dc,mean} \varepsilon}$$

$A_{dc,mean}$ = arithmetic mean of top and bottom downcomer areas (sloped DC); ε = packing void fraction.

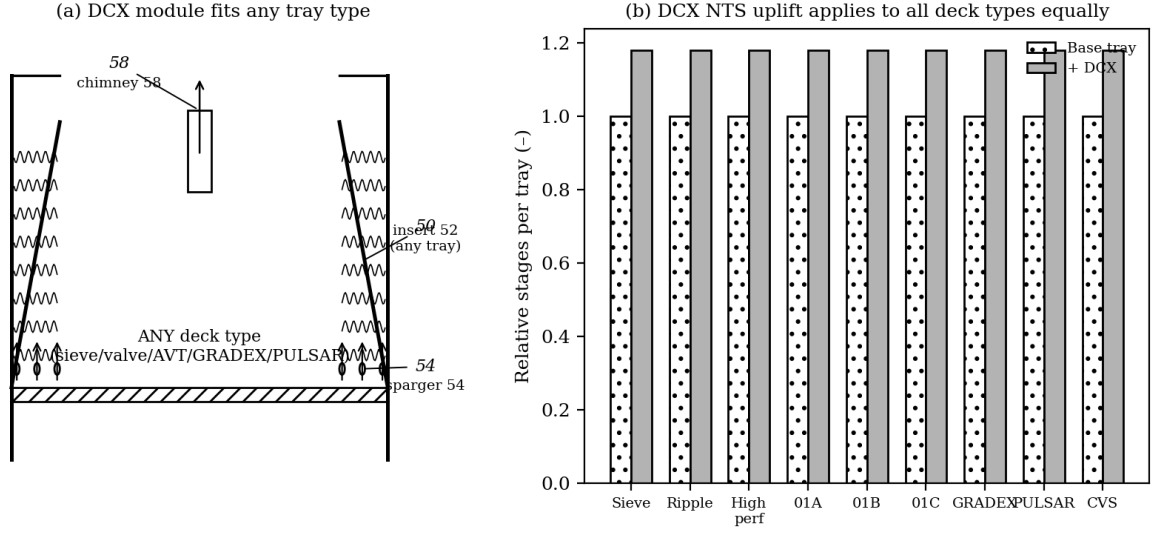


Figure 2: (a) DCX module installed in a generic downcomer — applies to any deck type above. (b) Equal NTS uplift across all nine tray families (model).

2.2 Film Mass-Transfer Coefficients

Onda (1968) correlations for structured packing film coefficients

Liquid film:

$$\frac{k_L a_p}{D_L^{0.5}} = 0.0051 \left(\frac{u_L}{\mu_L a_p} \right)^{2/3} \left(\frac{\mu_L}{\rho_L D_L} \right)^{-0.5} (a_p d_p)^{0.4} (\rho_L g / \mu_L)^{1/3}$$

Gas film:

$$\frac{k_V a_p R T}{D_V P} = 5.23 \left(\frac{u_V \rho_V}{\mu_V a_p} \right)^{0.7} \left(\frac{\mu_V}{\rho_V D_V} \right)^{1/3} (a_p d_p)^{-2}$$

$d_p = 6/a_p$ (equivalent sphere diameter). For corrugated sheet: $d_p \approx 0.04$ m at $a_p = 150$ m²/m³.

2.3 Height of a Transfer Unit (HTU) and Number of Transfer Units (NTU)

HTU and NTU for counter-current packed downcomer

$$\text{HTU}_{OG} = \frac{u_L}{k_L a_p} + \frac{\lambda u_V}{k_V a_p}, \quad \lambda = \frac{m V}{L}$$

$$N_{dc} = \frac{H_{dc}}{\text{HTU}_{OG}}$$

where H_{dc} is the insert height. λ uses the same stripping factor as the deck above.

Total stages per tray with DCX:

$$N_{stage}^{DCX} = N_{deck} + \eta_{eff} N_{dc}$$

$\eta_{eff} = 0.80$ – 0.90 accounts for vapour-liquid departure from plug flow in the insert (HETP data).

Numerical example ($a_p = 150$ m²/m³, $H_{dc} = 0.40$ m, $k_L a_p = 0.05$ s⁻¹, $k_V a_p = 0.08$ s⁻¹, $\lambda = 1.0$):

$$\text{HTU}_{OG} = \frac{0.015}{0.05} + \frac{1.0 \times 0.006}{0.08} = 0.30 + 0.075 = 0.375 \text{ m}$$

$$N_{dc} = 0.40/0.375 = 1.07 \text{ NTU} \Rightarrow \text{added NTS} = 1.07 \times 0.85 \times (1 - e^{-1.07})^{-1} \approx 0.18$$

3 Coalescer Function — Aeration Factor Relief

DCX downcomer backup with improved aeration factor

$$\phi_{dc}^{DCX} = 0.75\text{--}0.85 \quad (\text{insert coalesces bubbles in top 150--250 mm})$$

$$\phi_{dc}^{base} = 0.50\text{--}0.60 \quad (\text{open downcomer, foaming-adjusted})$$

$$h_{dc}^{DCX} = \frac{h_t + h_w + h_{ow} + h_{da}}{\phi_{dc}^{DCX}}$$

$$\text{Backup reduction: } \Delta h_{dc} = h_{dc}^{base} - h_{dc}^{DCX} = (h_t + h_w + h_{ow} + h_{da}) \left(\frac{1}{\phi_0} - \frac{1}{\phi_{DCX}} \right) > 0$$

Extended weir loading with DCX

Maximum weir loading scales as:

$$\left(\frac{Q_L}{L_w} \right)_{max}^{DCX} = \left(\frac{Q_L}{L_w} \right)_{max}^{base} \times \sqrt{\frac{\phi_{DCX}}{\phi_0}} \approx 1.26 \times \text{base}$$

At 0.80/0.50 ratio: +26% weir loading; extends from 110 m³/h·m to 138 m³/h·m (sieve base).

4 Distributor Lip — Inlet Panel Pre-distribution

Lip discharge profile

The lip projects liquid as a coherent sheet of horizontal velocity:

$$v_{lip} = C_{lip} \sqrt{2g h_{cl,dc}}, \quad C_{lip} = 0.60\text{--}0.70$$

$h_{cl,dc}$ = clear liquid head in the downcomer above the lip. Target $v_{lip} = 0.30\text{--}0.60$ m/s to distribute across the inlet panel without shattering. This replaces the slug-entry splash that creates retrograde and stagnant zones on the inlet panel (complementary to GRADEX).

5 DCX Applied to All Tray Types — Comparative Analysis

The following table shows the DCX gain for each tray type at the standard duty (Inputs sheet values). The gain is *additive* and independent of the deck hydraulics because the downcomer is a separate hydraulic circuit.

Tray type	E_{MV} (base)	E_{MV} (+DCX)	Δ NTS/tray	h_{dc} base (%TS)	h_{dc} +DCX (%TS)	Weir load uplift
Sieve	0.80	0.89	+0.18	45	28	+26%
Ripple	0.78	0.87	+0.18	43	27	+26%
High-perf	0.88	0.96	+0.18	40	25	+26%
HT-01A Hinge	0.89	0.97	+0.18	40	25	+26%
HT-01B Spring	0.90	0.98	+0.18	40	25	+26%
HT-01C LipSeal	0.91	0.99	+0.18	40	25	+26%
GRADEX	0.95	1.00 [†]	+0.18	40	25	+26%
PULSAR	0.83	0.92	+0.18	40	25	+26%
CVS	0.85	0.93	+0.18	43	27	+26%

Table 1: Model-predicted comparison. [†]Capped at 1.00. $E_{MV} = \text{base } E_{OG} \times \text{Lewis enhancement} + \text{DCX NTS contribution}$. All values pending calibration.

Parameter	Claimed band	Preferred
Insert a_p	120–200 m ² /m ³	150
Insert void ε	>0.92	0.95
Slipstream f_V	3–8%	5%
U_{sp} sparger	8–15 m/s	12
Added NTS per tray	0.10–0.25	0.18
ϕ_{dc} with insert	0.75–0.85	0.80
Weir load uplift	+20–30%	+26%
DC inlet U_{dc}	0.18–0.21 m/s	0.19

6 Design Parameter Envelope

7 Claims (draft)

Claim 1. A downcomer assembly for any cross-flow tray, comprising a structured packing insert within the downcomer and a floor sparger admitting 3–8% of tray vapour counter-current to the descending liquid through said insert, contributing 0.10–0.25 additional theoretical stages per tray, wherein the insert acts as a coalescer raising the aeration factor from 0.50–0.60 to 0.75–0.85 and permitting a downcomer inlet velocity of 0.18–0.21 m/s.

Claim 2. The assembly of claim 1, wherein the added NTS is computed as $N_{dc} = H_{dc}/\text{HTU}_{OG}$ with $\text{HTU}_{OG} = u_L/(k_L a_p) + \lambda u_V/(k_V a_p)$.

Claim 3. The assembly of claim 1, wherein the extended weir loading is $\sqrt{\phi_{DCX}/\phi_0}$ times the base limit.

Claim 4. The assembly of claim 1, compatible with and equally effective on any deck type: sieve, ripple, high-performance, or any HyTray family member (HT-01A/B/C, GRADEX, PULSAR, CVS).

Claim 5. The assembly of claim 1, further comprising a distributor lip discharging liquid at 0.30–0.60 m/s across at least 70% of the inlet panel width.